

WatchMan Database Schema

Core entities, recommendation data and relationships

Schema overview

The schema uses one unified content model for movies and TV/web series, plus user data, interaction data, recommendation persistence, semantic embeddings and ALS factors.

Core entities

Table/model	Purpose
users	User identity/account.
profiles	Profile information.
user_preferences	Favorite/disliked genre preferences.
contents	Unified movie + TV content and metadata.
genres / content_genres	Genre taxonomy and mapping.
languages / content_languages	Language taxonomy and mapping.
people / content_cast / content_crew	Cast and crew relationships.
content_external_ids	External identifiers.
content_videos	Trailer/video metadata.
saved_content	Saved/favorite relationship.
ratings	User-content ratings.
reviews	User reviews.
watch_history	Progress, completion and history.
interaction_events	Recommendation activity/event log.
content_embeddings	384-dimensional semantic vectors.
user_embeddings	384-dimensional user taste vectors.
als_user_factors	Versioned ALS user factors.
als_item_factors	Versioned ALS item factors.
recommendation_candidates	Candidate snapshots; audited as write-only.
recommendations	Persisted final score/rank/explanation.

Relationship map



Content embeddings and ALS factors are separate representations. pgvector is used for semantic vectors; ALS uses persisted latent factors. Final recommendation rows store ranking information while Redis provides runtime caching.

No clustering table is part of the current architecture.